Tan Dai Ngo

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Technical Skills

- **Programming:** Python, Java, SQL, MATLAB
- **Machine Learning:** scikit-learn, XGBoost, CatBoost, regression, classification, clustering, feature engineering, Optuna
- **Data & ML Systems:** Apache Spark, Kafka, Cassandra, Neo4j, MLflow, ETL pipelines, data processing, distributed systems
- **Deployment & GenAI:** FastAPI, Docker, AWS (EC2), Sentence Transformers, Qdrant, Retrieval-Augmented Generation (RAG), Ollama LLM

Education

Master of Engineering in Applied Data Science — **September 2025 - December 2027 (Expected)**
University of Victoria

- **GPA:** 8.68 / 9.00
- **Relevant Coursework:** Optimization for Machine Learning, Data Analysis and Pattern Recognition, Systems for Massive Datasets, Information Theory, Algorithms and Data Models, Data Mining, Applied Data Analysis

Master of Science in Computer Science — University of Chicago

Bachelor of Science in Economics — University of Washington

Experience

Software Developer | T-Mobile (via BeaconFire Inc.) — *Bellevue, WA, USA* | **June 2022 - April 2024**

- Improved reliability of operational data workflows by automating Kafka data reprocessing pipelines, reducing manual intervention by 80%.
- Reduced data retrieval latency for partner workflows to under 3 seconds by redesigning Cassandra schemas supporting 100k+ records.
- Enabled scalable processing of roaming and partner datasets by developing microservices supporting downstream analytics and operational systems.

Projects

Distributed Ocean Data ML Platform with RAG — Systems for Massive Datasets Project

- Built an end-to-end distributed ML pipeline using Apache Spark (Bronze → Silver → Gold architecture) to ingest and harmonize 10M+ NOAA and ONC ocean sensor observations for scalable feature engineering and time series forecasting.
- Developed and tuned XGBoost forecasting models using Optuna and MLflow, applying chronological holdout validation on temporally correlated ocean data (RMSE: 0.00586, R^2: 0.9999).
- Deployed FastAPI services for prediction and RAG-based querying using Sentence Transformers, Qdrant/FAISS, and Ollama, with retrieval benchmarking (hit@k: 1.0, term recall: 0.875), and a Streamlit dashboard for interactive exploration.

End-to-End Machine Learning System for Multi-Output Fuel Blending — Shell.ai Hackathon 2025

- Built and deployed a production ML system for multi-output regression across 10 chemical blend properties using FastAPI, Docker, and AWS EC2 for real-time inference.
- Benchmarked XGBoost and CatBoost using 5-fold cross-validation, selecting CatBoost after achieving significantly lower error (MAPE: 0.64 vs. 1.29).
- Designed feature engineering and inference pipelines using weighted aggregation, mixture metrics based on entropy, and serialized preprocessing to ensure consistent production predictions.

Stock Sentiment Analysis (Team of 4) — Algorithms and Data Models Project

- Built a sentiment analysis pipeline over 15k+ tweets across 10 tech stocks, enabling large-scale market sentiment analysis, by integrating X API data, SQLite/Neo4j storage, and FinTwitBERT modeling.
- Identified measurable sentiment-return relationships, including statistically significant predictive effects for NVIDIA, by implementing engagement-weighted sentiment aggregation and Granger causality testing.
- Analyzed equity volatility dynamics across major tech stocks, revealing persistent volatility clustering, by applying GARCH(1,1) modeling and automated cross-stock analysis workflows.